

PART B: PLATFORM PLAN

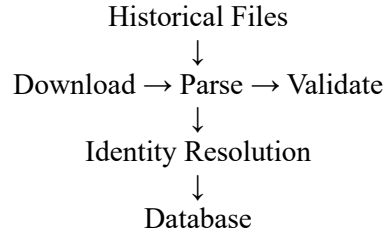
1.Ingestion

- NSE and BSE are handled as **separate ingestion pipelines** because their file formats and schedules differ.
- A scheduled worker checks for the daily file, downloads it, and stores the **original file unchanged** in object storage.
- Each exchange has a versioned **parser/adaptor** that converts data into one common schema.
- Validate columns, dates, numbers, duplicates, row counts, and source totals before publishing.
- **Holiday**: use a trading calendar; don't treat it as missing data.
- **Late file**: mark PENDING, retry, then alert if still unavailable.
- **Never arrives**: mark PARTIAL, keep last valid data, alert, and allow later replay.
- **Format change**: quarantine the file → alert → update parser → replay and validate

NSE/BSE → Fetch → Parse → Validate → Store → Publish.

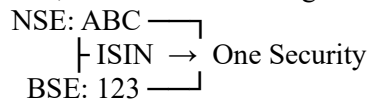
2.Backfill

- Collect approximately 9 years / 2,200 trading days of NSE and BSE files.
- Download and process files using bounded parallel workers.
- Store originals, parse, validate, resolve identity, and load into the database.
- Use checksum + date + exchange + parser version to prevent duplicates.
- Approximately 4,400 files can be processed in around 1–2 hours, depending on network, exchange throttling, file size, and database performance



3.Identity

- The same company can have different NSE/BSE symbols and names.
- Use ISIN as the primary identifier.
- Maintain a security master containing ISIN, exchange, symbol, scrip code, company name, and validity dates.
- If ISIN is unavailable, use exchange identifiers.
- Never merge companies only by name; unresolved records go for manual mapping.



4.Storage & Serving

Schema: Blueprint of Database

security

- isin
- exchange
- symbol
- scrip_code
- company_name
- valid_from
- valid_to

mtf_daily

- trading_date
- exchange
- isin
- symbol
- amount_financed
- fresh_exposure
- exposure_liquidated
- securities_count

5. Correctness

- Validate at **file, row, aggregate, and time-series levels.**

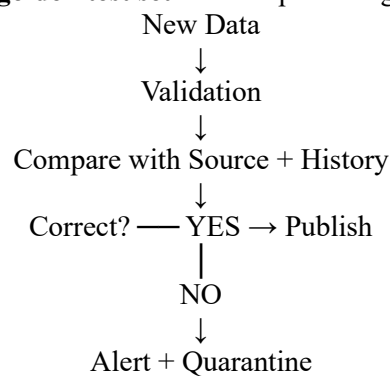
Check:

Required columns
Valid dates/numbers
Duplicates/nulls
Source totals
Unexpected zero values
Unusual daily movements

Monitor:

Last successful ingestion
Last received file
Row count
Total MTF value
Validation status
Processing time

- Alert on missing/late files, schema changes, duplicates, stale data, and suspicious movements.
- Keep real exchange files as a **golden test set** to catch parser regressions.



6. Roadmap & Cost

Timeline	What ships
2 Weeks	NSE/BSE adapters, raw storage, database schema, validation, aggregates, JSON/API, dashboard, freshness monitoring
1 Month	9-year backfill, ISIN security master, historical stock data, retry/replay
2 Months	Data-health dashboard, alerts, parser regression tests
3 Months	Admin tools, full monitoring, production hardening, advanced analytics

Estimated Cost

~\$20–70/month for a low-volume deployment:

- Object storage: \$1–5
- Worker: \$5–15
- PostgreSQL: \$10–30
- CDN/API caching: \$0–10
- Monitoring: \$0–10

Cost increases with higher traffic, larger databases, high availability, and more monitoring.